

Implementation of Radix-2 8-Point FFT with Streaming Input Using Verilog

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1 Objective

- To implement a Radix-2 8-point FFT using a pipelined butterfly architecture in Verilog.
- To support streaming input where one complex sample $8 - \text{bitreal} + 8 - \text{bitimaginary}$ is received per clock cycle.
- To verify results with MATLAB and Intel FFT IP.
- To analyze throughput, latency, and F_{max} .

2 Theory

2.1 Discrete Fourier Transform

The N -point DFT is given by:

$$X_k = \sum_{n=0}^{N-1} x_n e^{-j \frac{2\pi}{N} kn} \quad (1)$$

2.2 Radix-2 FFT

Radix-2 FFT reduces computational complexity from $\mathcal{O}N^2$ to $\mathcal{O}N \log N$ using divide-and-conquer.

For $N = 8$:

- Number of stages $= \log_2 8 = 3$
- Each stage consists of butterfly operations

2.3 Butterfly Operation

A butterfly computes:

$$Y_0 = A + B \cdot W_N^k \quad (2)$$

$$Y_1 = A - B \cdot W_N^k \quad (3)$$

where $W_N^k = e^{-j \frac{2\pi k}{N}}$

3 Streaming FFT Architecture

3.1 Concept

- One complex sample arrives per clock cycle
- Total 8 cycles required to input full frame
- Pipeline stages process data simultaneously

3.2 Pipeline Design

- Stage 1: Distance = 4
- Stage 2: Distance = 2
- Stage 3: Distance = 1

4 Verilog Implementation

4.1 Design Features

- Fixed-point representation: $Q(5,2) + 1$ sign bit
- Signed arithmetic
- Pipelined registers between stages
- Streaming input interface

4.2 Modules Used

- Butterfly Unit
- Twiddle Factor Multiplication
- Pipeline Registers
- Top FFT Module

5 Testbench

- Inputs provided sequentially over 8 clock cycles
- Compared output with MATLAB FFT results
- Verified correctness for multiple test vectors

6 MATLAB Verification

```
X(0) = 164.0000 + j8.0000
X(1) = -4.0000 + j9.6569
X(2) = -4.0000 + j4.0000
X(3) = -4.0000 + j1.6569
X(4) = -4.0000 + j0.0000
X(5) = -4.0000 - j1.6569
X(6) = -4.0000 - j4.0000
X(7) = -4.0000 - j9.6569
```

Figure 1: matlab results

7 Results

```
# Time=555000 | OUT_real = 164.0000 | OUT_imag = 8.0000
# Time=565000 | OUT_real = -4.0000 | OUT_imag = 9.5000
# Time=575000 | OUT_real = -4.0000 | OUT_imag = 4.0000
# Time=585000 | OUT_real = -4.0000 | OUT_imag = 1.5000
# Time=595000 | OUT_real = -4.0000 | OUT_imag = 0.0000
# Time=605000 | OUT_real = -4.0000 | OUT_imag = -1.5000
# Time=615000 | OUT_real = -4.0000 | OUT_imag = -4.0000
# Time=625000 | OUT_real = -4.0000 | OUT_imag = -9.5000
```

Figure 2: fft pipeline results

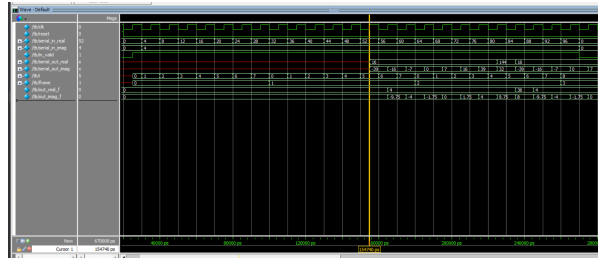


Figure 3: waveform of fftpipe, we can observe latency of 12 cycles

7.1 Comparison

- Real and imaginary outputs matched with small quantization error

8 Resource Utilization and Fmax

Flow Summary		
Flow Status	Successful - Sun Apr 12 19:36:54 2026	
Quartus Prime Version	20.1.0 Build 711 06/05/2020 SJ Lite Edition	
Revision Name	fftfixed	
Top-level Entity Name	top	
Family	Arria II GX	
Logic utilization	3 %	
Total registers	928	
Total pins	41 / 176 (23 %)	
Total virtual pins	0	
Total block memory bits	0 / 2,039,904 (0 %)	
DSP block 18-bit elements	0 / 232 (0 %)	
Total GXB Receiver Channel PCS	0 / 4 (0 %)	
Total GXB Receiver Channel PMA	0 / 4 (0 %)	
Total GXB Transmitter Channel PCS	0 / 4 (0 %)	
Total GXB Transmitter Channel PMA	0 / 4 (0 %)	
Total PLLs	0 / 4 (0 %)	
Total DLLs	0 / 2 (0 %)	
Device	EP2AGX45CU1713	
Timing Models	Final	

Figure 4: Resouce utilisation of FFT pipe

Compilation Report - ffftfixed

Table of Contents

- Flow Summary
- Flow Settings
- Flow Non-Default Global Set
- Flow Elapsed Time
- Flow OS Summary
- Flow Log

Slow 900mV 100C Model Fmax Summary

<< Filter >>

	Fmax	Restricted Fmax	Clock Name	Note
1	127.96 MHz	127.96 MHz	clk	

Figure 5: Fmax of FFT pipe

Flow Summary		
Flow Status	EDA Netlist Writer Failed - Sun Apr 12 20:03:08 2026	
Quartus Prime Version	20.1.0 Build 711 06/05/2020 SJ Lite Edition	
Revision Name	fftp1	
Top-level Entity Name	suraj	
Family	Arria II GX	
Logic utilization	4 %	
Total registers	1092	
Total pins	61 / 176 (35 %)	
Total virtual pins	0	
Total block memory bits	1,096 / 2,039,904 (< 1 %)	
DSP block 18-bit elements	4 / 232 (2 %)	
Total GXB Receiver Channel PCS	0 / 4 (0 %)	
Total GXB Receiver Channel PMA	0 / 4 (0 %)	
Total GXB Transmitter Channel PCS	0 / 4 (0 %)	
Total GXB Transmitter Channel PMA	0 / 4 (0 %)	
Total PLLs	0 / 4 (0 %)	
Total DLLs	0 / 2 (0 %)	
Device	EP2AGX45CU1713	
Timing Models	Final	

Figure 6: Resouce utilisation of FFT IP

Compilation Report - fftp1

Table of Contents

- Flow Summary
- Flow Settings
 - Flow Non-Default Global Settings
 - Flow Elapsed Time
 - Flow OS Summary
 - Flow Log
- Analysis & Synthesis
- Filter
- Assembler
- Timing Analyzer
 - Summary
 - Parallel Compilation
 - SDC File List
 - Clocks
 - Clock0(900mV 100C Model)

Slow 900mV 100C Model Fmax Summary

	Fmax	Restricted Fmax	Clock Name	Note
1	234.58 MHz	234.58 MHz	clk	

Figure 7: Fmax of FFT IP